Commutator

Def. Let G be a group. Given $x, y \in G$, define

$$[x, y] \overset{\text{def}}{=} x^{-1}y^{-1}xy$$

is called the commutator of x and y. And, given two subset A and B,

$$[A, B] \overset{\text{def}}{=} \langle [a, b] \mid a \in A,\ b \in B \rangle$$

Particularly, $G' \overset{\text{def}}{=} [G, G]$ is called the derived subgroup.

Prop. G' char G, (So $G' \trianglelefteq G$).

Proof. Given $\sigma \in \mathrm{Aut}(G)$ and $[x, y]$ for arbitrary $x, y \in G$,

$$\sigma([x, y]) = \sigma(x^{-1}y^{-1}xy) = \sigma(x)^{-1} \cdot \sigma(y)^{-1} \cdot \sigma(x) \cdot \sigma(y) = [\sigma(x), \sigma(y)]$$

So, $\sigma[[G, G]] = [G, G]$.

Prop. Given $H \leq G$, $H \trianglelefteq G$ if and only if $[H, G] \leq H$

Proof. $H \trianglelefteq G \iff$ for all $g \in G$ and $h \in H$, $ghg^{-1} \in H$

$[H, G] \leq H \iff$ for all $g \in G$ and $h \in H$, $[h, g] = h^{-1}g^{-1}hg \in H$.

So proved. ▨

Prop. Given $N \trianglelefteq G$, G/N is abelian if and only if $G' \leq N$.

Proof. G/N is abelian iff for any $xN, yN \in G/N$, $xyN = yxN$

iff for any $x, y \in G$, $x^{-1}y^{-1}xy \in N$

iff $G' \leq N$. ▨

Note: G' is the smallest normal subgroup the quotient is abelian. In other say, G/G' is the largest abelian quotient group.

$$G' = \bigcap_{\substack{N \trianglelefteq G \\ G/N \text{ is abelian}}} N$$

Prop. If $\varphi : G \to A$ is a Homomorphism into an abelian group A, then $G' \leq \ker \varphi$.

Proof. By the first Isomorphism Theorem, $G/\ker \varphi \cong \mathrm{Im}\,\varphi \leq A$. Since A is abelian, so $\mathrm{Im}\,\varphi$ is also abelian. Therefore, $G' \leq \ker \varphi$. ▨

Alternatively, given $x, y \in G$, $\varphi([x, y]) = \varphi(x)^{-1}\varphi(y)\varphi(x)\varphi(y) = 1_A$. ▨

Homomorphism decomposition.

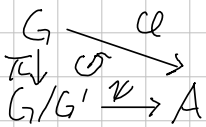
Let $\varphi : G \to A$ be a group homomorphism into abelian group A.

By the first-Isomorphism theorem, $G/\ker\varphi \cong \operatorname{Im}\varphi \leq A$, so $G' \leq \ker\varphi$.

Therefore, $\psi : G/G' \to A ; aG' \mapsto \varphi(a)$ is well-defined, because

$$a_1G' = a_2G' \Leftrightarrow a_1^{-1}a_2 \in G' \Rightarrow a_1^{-1}a_2 \in \ker\varphi \Leftrightarrow \varphi(a_1^{-1}a_2) = e \Leftrightarrow \varphi(a_1) = \varphi(a_2)$$

Now, the onto homomorphism, $\pi : G \to G/G' ; g \mapsto gG'$ satisfies :



$\varphi = \psi \circ \pi$.

And,

$$\ker\varphi = \varphi^{-1}[\{e\}] = \pi^{-1}[\psi^{-1}[\{e\}]] = \pi^{-1}[\ker\psi].$$

Commutator of Typical Subgroups.

D_{2n}, the dihedral of order $2n$.

Prop. $D_{2n}' = \langle r^2 \rangle \cong \begin{cases} C_n & \text{if } n \text{ odd} \\ C_{\frac{n}{2}} & \text{if } n \text{ even} \end{cases}$

S_n, the symmetric group on n letters

Prop. $S_n' = A_n$.
Proof. Since $S_n/A_n \cong C_2$, abelian, so $S_n' \leq A_n$.

Pf of $S_3' = A_3$:
$[S_3, S_3] = \langle [a,b] \mid a, b \in S_3 \rangle$.
$[(12),(13)] = (12)(13)(12)(13) = (132)(132) = (123)$
$[(12),(23)] = (12)(23)(12)(23) = (231)(231) = (132)$
$[(13),(23)] = (13)(23)(13)(23) = (321)(321) = (123)$
$[(12),(123)] = (12)(123)^{-1}(12)(123) = (12)(321)(12)(123) = (132)$
$[(13),(123)] = (13)(123)^{-1}(13)(123) = (13)(321)(13)(123) = (132)$
$[(23),(123)] = (23)(123)^{-1}(23)(123) = (23)(321)(23)(123) = (132)$.
$[(123),(132)] = id$.

Therefore, $[S_3, S_3] = \langle (123), (132) \rangle = \{id, (123), (132)\} = A_3$.

Alternatively, Since A_3 is normal subgroup of S_3 and $S_3/A_4 \cong C_2$, so $[S_3, S_3] \subseteq A_3$. Conversely, since $(132) \in A_3$ and $(132) = [(12),(23)] \in [S_3, S_3]$.
Therefore $[S_3, S_3] \geq A_3 = \langle (132) \rangle$.

Since $S_n = \langle (12), (123\cdots n) \rangle$, because:
$(12\cdots n)(12)(n\cdots 21) = (23)$
$(12\cdots n)(23)(n\cdots 21) = (34)$
$\vdots$
$(12\cdots n)(n-2\ n-1)(n\cdots 21) = (n-1\ n)$
and for any $1 \leq i \leq n$
$(i\ i+1)(i+1\ i+2)(i\ i+1) = (i\ i+2)$
$(i\ i+2)(i+2\ i+3)(i\ i+2) = (i\ i+3)$
$\vdots$
$(i\ i+\delta)(i+\delta\ i+\delta+1)(i\ i+\delta) = (i\ i+\delta)$, inductively,
$\langle (12), (12\cdots n) \rangle$ contains all transpositions in S_n, so it generates S_n.
Now,
$S_n' = [S_n, S_n] =$

A_n, the alternating group.

Q8. The Quaternion.